

Ultrasound-Driven Passive Haptic Actuator Based on Amplifying Radiation Force Using Simple Lever Mechanism

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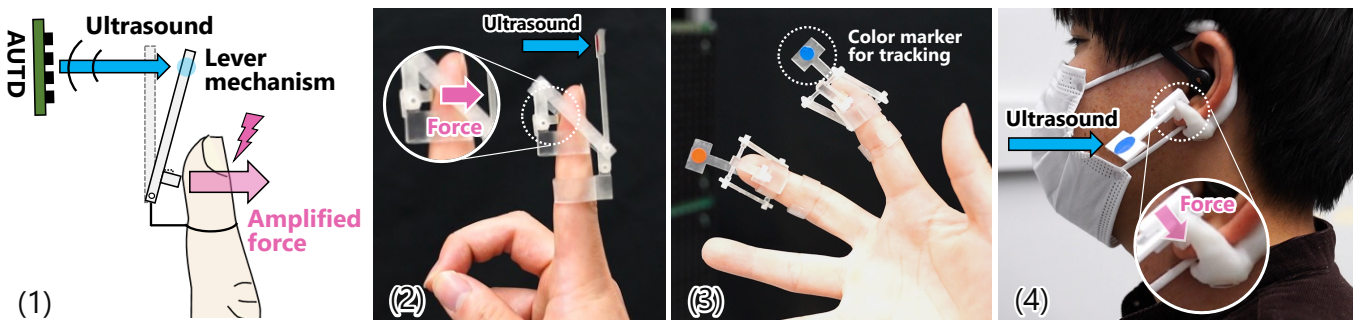


Figure 1: 1) Concept of the ultrasound-driven passive haptic actuator, 2) finger-mounted passive haptic device, 3) double haptic device and color marker, and 4) earring-type haptic device presenting stimulus to earlobe.

ABSTRACT

Haptics is a promising modality, which realizes intuitive human motion guidance and an immersive game experience. For a natural tactile experience, a lightweight and powerful wearable haptic device is required. In this study, we develop a lightweight passive haptic device (6.2 g) remotely driven by airborne ultrasound. This device can present a strong haptic stimulus of 400 mN (40 gf) by amplifying the applied ultrasound acoustic radiation force 19.6 times using a simple lever mechanism. Moreover, since the radiation force is presented at the sound velocity, the presentation speed of the amplified force is still high. In this demo, participants can experience a lightweight passive haptic actuator worn on their fingertips. This device can present a static force of 400 mN and low-frequency vibration in 45 ms. We also demonstrate an earring-type passive haptic device, which presents a haptic stimulus to the earlobe.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Passive haptic actuator, ultrasound, lever mechanism

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1 INTRODUCTION

Haptics is a promising modality, which has been used in many applications such as human motion guidance [Spiers and Dollar 2016] and improving the sense of immersion in VR space. To present tactile stimulus, users must touch with mechanical devices and it restricts the user's natural movement. There is a need for a low-burden haptic display that does not restrict the user's movement.

As a low-burden tactile display, airborne ultrasound tactile display (AUTD) has been proposed [Carter et al. 2013; Hoshi et al. 2010]. AUTD is the device in which ultrasound transducers are arranged in an array. By focusing ultrasound using an AUTD, a noncontact force called acoustic radiation force is generated, which delivers a tactile sensation to human skin.

However, the presentable sensation and the workspace of AUTD are limited. Since the ultrasound radiation force is weak (several tens mN), AUTD cannot present a strong haptic sensation (e.g., reaction force when touching a hard object). Long-distance tactile presentation is also difficult for AUTD. When focusing ultrasound

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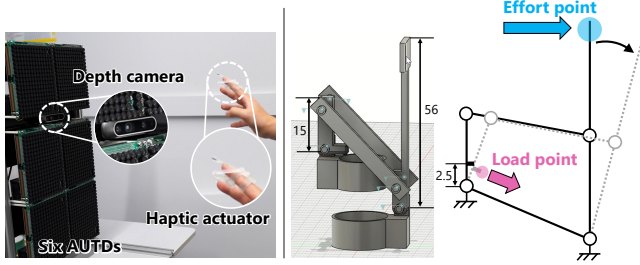


Figure 2: L) System overview, R) schematic of developed finger-mounted passive haptic device. Dimensions are in mm.

at a distant point from the AUTD, the sound pressure attenuation and the focus blur is enlarged [Carter et al. 2013; Hasegawa and Shinoda 2018]. The practical presentation distance is comparable to the aperture size of the AUTD.

In this study, we propose to amplify the ultrasound radiation force using a simple lever mechanism to present a strong haptic stimulus while keeping the user's burden low (Fig. 1-(1)). This lever-based amplifier is, in other words, a passive haptic actuator that is remotely driven by airborne ultrasound. We preliminary confirmed that the prototype of the finger-mounted passive haptic device (Fig. 1-(2)) presents a strong haptic stimulus of 400 mN (40 gf). Even when the actuator is located beyond the aperture size of the AUTD, the amplified haptic stimulus can be clearly perceived. Lu et al. proposed a passive haptic device using magnetic force, but its workspace is limited to the inside of the cylindrical electro-magnet [Lu et al. 2022]. We believe that the burden of wearing the actuator is low because the passive actuator is lightweight (6.2 g). Moreover, even if multiple actuators are simultaneously worn (as shown in Fig. 1-(3)), complicated wiring is not required.

A lever mechanism is efficient for amplifying the ultrasound radiation force. Since the radiation force is presented at the sound velocity, the force can rotate a lightweight lever at high speed. Therefore, the amplified force is presented fast enough for tactile presentation. For example, if ultrasound force rotates the effort point of a lever at 170 m/s (half the sound velocity), even if the amplification factor is 20 times, the presentation velocity of the amplified force is still high, $170 / 20 = 8.5$ m/s.

In this demo, users can experience an ultrasound-driven passive haptic device that is worn at their fingertips. This actuator presents a strong force (400 mN) and a vibrotactile sensation. As an application, we also demonstrate an earring-type passive haptic device.

2 PROTOTYPE SYSTEM

2.1 Overview

The developed ultrasound-driven haptic actuator and its drive system are shown in Fig. 2-L. The actuator shown in Fig. 1-(2) presents a haptic stimulus to a fingertip. The drive system consists of the six AUTDs and a depth camera (Intel Real Sense D435). One AUTD unit we used has 249 ultrasound transducers driving at 40 kHz [Suzuki et al. 2021]. The depth camera measures the 3D position of the color marker attached to the tip of the actuator (lever part) and the

AUTDs present an ultrasound focus to the detected position. The focus position is changed in real-time at 60 fps.

2.2 Passive Haptic Actuator

The developed haptic actuator for fingertip is shown in Fig. 1-(2) and its schematic is shown in Fig. 2-R. This actuator was developed using a lever-crank mechanism. By presenting an ultrasound focus to the tip of this actuator (the tip of the lever), the amplified radiation force is finally applied to the fingertip. We designed this device to have an amplification factor of 20. The actuator is made of plastic and weighs 6.2 g.

The passive actuator can present static force and vibrotactile stimulus. If the input radiation force is constant, static force is presented. By modulating the input radiation force at 5-10 Hz, a vibrotactile stimulus is presented to the fingertips according to that frequency. Multiple actuators (Fig. 1-(3)) can also be driven simultaneously by a time-division presentation of ultrasound focus.

2.3 Preliminary Evaluation

We preliminarily confirmed that the passive haptic actuator can present a strong haptic stimulus of 400 mN (40 gf) as fast as 45 ms. In the experiment, the haptic actuator was driven at a distance of 30 cm from the six AUTDs and the amplified force was measured using a force gauge (IMADA ZTS-50N). The input radiation force was 21 mN thus, the amplification factor is 19.6. The factor is consistent with the theoretical value (20).

3 POTENTIAL APPLICATION

The actuator can be applied to a jewelry-shaped passive haptic device. Since jewelry is worn in daily life, the lightweight jewelry-shaped device can provide a low-burden and natural tactile experience. In this demo, we show an earring-type passive haptic device that presents vibration and static force to the earlobe (Fig. 1-(4)). The device weighs 14 g and can be worn with a low burden. We believe that our passive actuator opens the door to a more natural tactile experience.

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REFERENCES

- Tom Carter, Sue Ann Seah, Benjamin Long, Bruce Drinkwater, and Sriram Subramanian. 2013. UltraHaptics: multi-point mid-air haptic feedback for touch surfaces. In *Proceedings of the 26th annual ACM symposium on User interface software and technology*. ACM, 505–514.
- Keisuke Hasegawa and Hiroyuki Shinoda. 2018. Aerial vibrotactile display based on multiunit ultrasound phased array. *IEEE transactions on haptics* 11, 3 (2018), 367–377.
- Takayuki Hoshi, Masafumi Takahashi, Takayuki Iwamoto, and Hiroyuki Shinoda. 2010. Noncontact tactile display based on radiation pressure of airborne ultrasound. *IEEE Transactions on Haptics* 3, 3 (2010), 155–165.
- Xiong Lu, Yuxing Yan, Beibei Qi, Huang Qian, Junbin Sun, and Aaron Quigley. 2022. Contactless haptic display through magnetic field control. *IEEE Transactions on Haptics* (2022).
- Adam J Spiers and Aaron M Dollar. 2016. Design and evaluation of shape-changing haptic interfaces for pedestrian navigation assistance. *IEEE transactions on haptics* 10, 1 (2016), 17–28.
- Shun Suzuki, Seki Inoue, Masahiro Fujiwara, Yasutoshi Makino, and Hiroyuki Shinoda. 2021. AUTD3: Scalable airborne ultrasound tactile display. *IEEE Transactions on Haptics* 14, 4 (2021), 740–749.